Methodoloy:

1. Get acceleration data in csv / excel form
2. Use numerical intergration on discrete data points, since data was collected at specific, dicrete time inervals rather than as a continuous mathematical function
3. Highly Reccomended method is the Trapezoidal Rule:
    - Most practical methd for experimental data like motion.
    - It assumes the acceleration changes linearly between the pairs of data points to form a trapezoid.
    - We use it cause it handles variable time steps, which can happen in sensor data if a frame drops, plus it it easy to implement in python
4. The formula:
$$v_{n+1} = v_n + \frac{a_n + a_{n+1}}{2} * \delta t$$
    - a is acceleration, v is velocity δt is the time difference $t_{n+1} - t_n$

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import cumulative_trapezoid

df = pd.read_csv("Raw Data.csv")
```

```
t = df['Time (s)'].to_numpy()
a = df['Linear Acceleration y (m/s^2)'].to_numpy()

v = cumulative_trapezoid(a, t, initial=0)

x = cumulative_trapezoid(v, t, initial=0)

df['Velocity (m/s)'] = v
```

```
df['Position (m)'] = x
df.to_csv("Skateboard Results.csv", index=False)

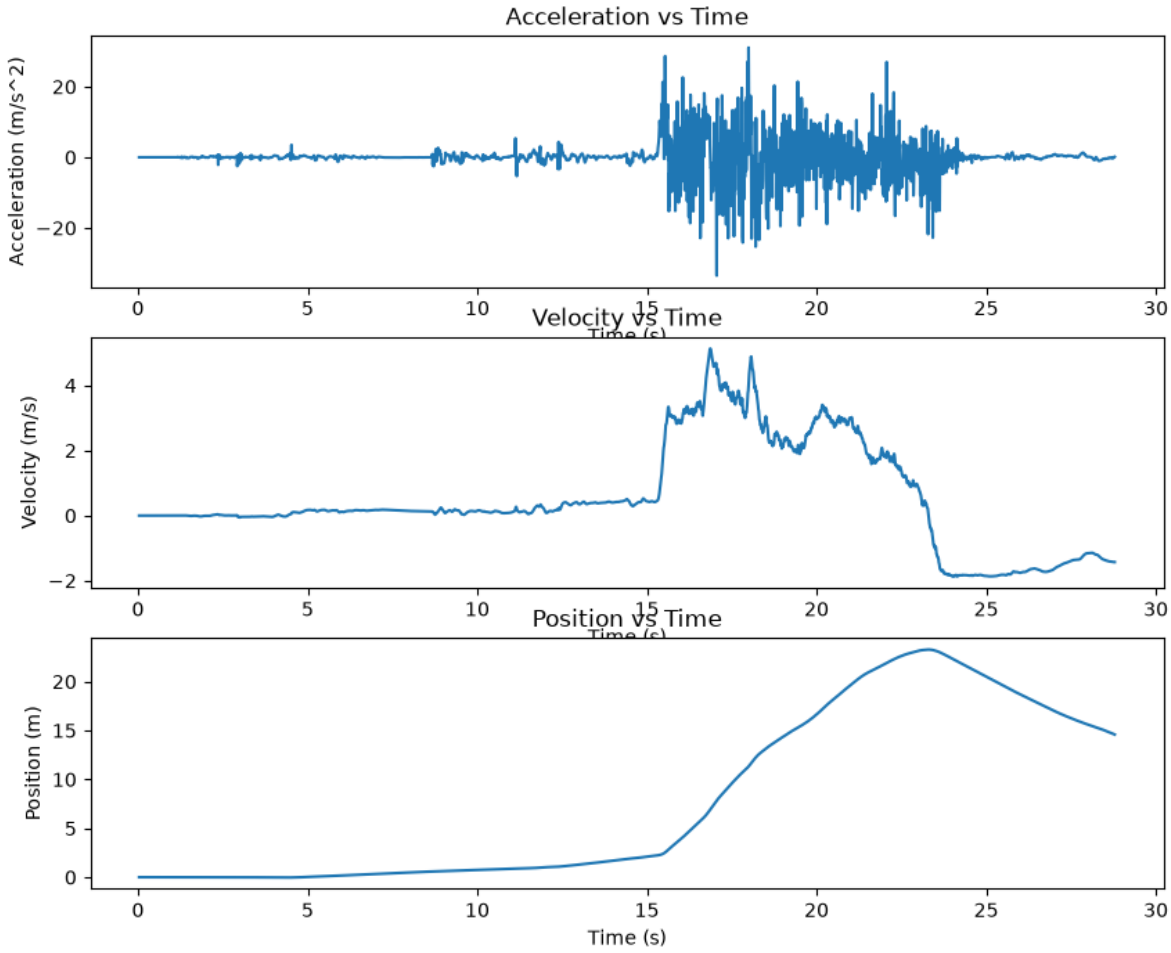
plt.figure(figsize=(10, 8))
plt.subplot(3, 1, 1)
plt.plot(t, a)
plt.xlabel('Time (s)')
plt.ylabel('Acceleration (m/s^2)')
plt.title('Acceleration vs Time')

plt.subplot(3, 1, 2)
plt.plot(t, v)
plt.xlabel('Time (s)')
plt.ylabel('Velocity (m/s)')
plt.title('Velocity vs Time')

plt.subplot(3, 1, 3)
plt.plot(t, x)
plt.xlabel('Time (s)')
plt.ylabel('Position (m)')
plt.title('Position vs Time')


# # print  ( df['Time (s)'].head(10))
```

```
Text(0.5, 1.0, 'Position vs Time')
```


Acceleration vs Time
Acceleration (m/s^2)
20
0
−20
0
5
10
15
20
25
30
Velocity vs Time
Velocity (m/s)
4
2
0
−2
Position vs Time
Position (m)
20
15
10
5
0
Time (s)